

The Global Race for Talent: Brain Drain, Knowledge Transfer and Growth

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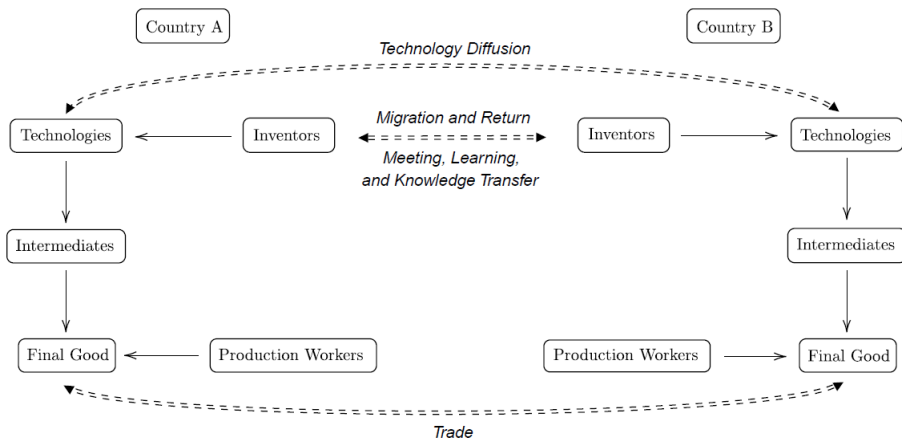
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3. Migrants on average increase productivity after migration
 - File on average 0.86 more patents per year
 - Increase patenting by 42% compared to sample mean
4. Locals tend to become more productive after co-inventor emigrates
 - File on average 0.366 more patents per year
 - Increase patenting by 18% compared to sample mean

Model Summary

- OLG model in GE with innovation-based endogenous growth in discrete time



- New technology $\uparrow$ quality of intermediary goods and $\uparrow$ productivity growth

Model - Setup

- Unit mass of individuals of each nationality $c \in \{A, B\}$
- Survival with probability δ each period
 - replaced by newborn when die
- Exogenous occupation: inventors I_c or production workers L_c
 - $L_c + I_c = 1$
- Inventors can move across countries
 - Sum of local inventors in country A : $\underbrace{\mu_{AA}}_{\text{local inventors}} + \underbrace{\mu_{BA}}_{\text{migrant inventors}} = I_A$

Model - Inventors

- Born with **talent** $z \sim$ Pareto distribution $\tilde{F}_c(z)$ with scale 1 and shape θ_c
 - z increases endogenously by learning from other inventors
- Every period, draw idiosyncratic **foreign productivity differential** ϵ
 - $\epsilon \sim \text{AR}(1)$ with persistence ρ_c , mean 0 and variance $\frac{\omega_c^2}{1-\rho_c^2}$
 - idiosyncratic reasons for being more productive abroad

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 - idiosyncratic reasons for being more productive abroad
- Inventors produce bundles of technologies $q_t \geq 1$:

$$q_t(z_t, \epsilon_t) = \begin{cases} z_t & \text{if local (living in country of origin)} \\ \max\{z_t + \epsilon_t, 1\} & \text{if migrant (living abroad)} \end{cases} \quad (1)$$

- $q \sim F_{j,t}(q)$ endg distribution produced by inventor type $j \in \{AA, AB, BB, BA\}$
- $F_c(q)$ endogenous distribution produced in country c

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- Probability $(1 - \lambda)$ do not meet anyone $\implies$ no learning, same z
- Individuals with higher z learn relatively more $\partial^2 z_t / \partial z_{t-1} \partial \hat{q}_{t-1} > 0$

Model - Meeting Probability

- Every inventor can meet any of the 4 types of inventors $j \in \mathcal{J} = \{AA, AB, BB, BA\}$
- Conditional on meeting, probability of type i meeting an inventor of type j :

$$\psi_{i,j,t} = \frac{\mu_{j,t}}{\sum_{j' \in \mathcal{J}} \mu_{j',t}} \tilde{\xi}_{i,j}$$

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- Frictionless meetings when $\tilde{\xi}_{i,j} = 1$
- Probabilities sum to 1 $\sum_{j \in \mathcal{J}} \psi_{i,j} = 1 \quad \forall i$

Model - Production of Goods

- Final good Y_c is numeraire and competitively produced:

$$Y_{c,t} = \frac{1}{1-\alpha} (L_c)^\alpha \int_0^1 (A_{j,c,t})^\alpha (k_{j,c,t})^{1-\alpha} dj$$

- L_c labor workers
- $k_{j,c,t}$ intermediate goods
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- L_c labor workers
 - $k_{j,c,t}$ intermediate goods
 - $A_{j,c,t}$ quality of intermediate
- Intermediate input k_j demand curve

$$P_{j,c} = (L_c)^\alpha (A_{j,c})^\alpha (k_{j,c})^{-\alpha} \quad (2)$$

- Monopolist produces each intermediate good
- Assume $MC = 1 - \alpha$
- Maximizing profits gives: $\Pi_{j,c} = \alpha L_c A_{j,c}$

Model - Aggregate Productivity, Output and Wage

- Aggregate productivity $\bar{A}_c$ equals average quality of intermediate goods $A_{j,c}$
- Aggregate output in equilibrium

$$Y_c = \frac{1}{1-\alpha} L_c \bar{A}_c \quad (3)$$

- Equilibrium worker wage

$$w_c = \frac{\alpha}{1-\alpha} \bar{A}_c \quad (4)$$

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2. Exogenous technology spillovers

- From frontier economy to laggard at rate σ , at no cost (ex: copying)
- New quality increase of:

$$\tilde{\sigma}_{c,t} = \sigma \max\{\bar{A}_{-c,t} - \bar{A}_{c,t}, 0\}$$

Model - Value Function for Intermediary Good Firms

- Value of owning intermediate good product line with quality $A_{j,c,t}$

$$J(A_{j,c,t}) = \Pi_{j,c,t} + \frac{1}{1+r} \left[x_{c,t} \left(\int_1^\infty (J(A_{j,c,t} + q\bar{A}_{c,t} + \tilde{\sigma}_{c,t}, t+1) - p_{j,c,t}(q)) dF_c(q) \right) + (1 - x_{c,t}) J(A_{j,c,t+1} + \tilde{\sigma}_{c,t}, t+1) \right]$$

- Assume inventors get all surplus from technology purchase

- Profits of inventor with talent z : $\pi_c(z, t) = \underbrace{(\mu_{Ac} + \mu_{Bc})^{\nu-1}}_{\text{prob match with firm}} \times \underbrace{p_{c,t}(q(z))}_{\text{revenues from } q}$

Timing - Inventors Problem

Every period, inventors of type AA:

1. Draw ϵ , foreign productivity shock
2. Meet with other inventors and (potentially) increase talent z
3. Decide to migrate or not $W_{AA}(z, \epsilon, t)$
4. Compute continuation value $V_{AA}(z, \epsilon, t)$
5. Produce technology bundle q
6. Sell technology to intermediate good producers, if match to firm
7. Survive to next period with probability δ

Model - Migration Decision

- Every period, inventors decide if they want to move
 - Fixed migration cost: $\kappa \bar{A}_{c,t}$
 - WLOG, no cost to return to homeland; can emigrate again
- Migration decision for local inventor in country A:

$$W_{AA}(z, \epsilon, t) = \max\{V_{AA}(z, \epsilon, t), V_{AB}(z, \epsilon, t) - \kappa \bar{A}_{c,t}\} \quad (5)$$

- Return migration decision for migrant inventor living in country B

$$W_{AB}(z, \epsilon, t) = \max\{V_{AB}(z, \epsilon, t), V_{AA}(z, \epsilon, t)\} \quad (6)$$

Model - Value Functions for Inventors

- Locals from country A living in A

$$V_{AA}(z, \epsilon, t) = \pi_A(z, t) + \beta\delta \int_{-\infty}^{\infty} \left(\lambda \sum_j \psi_{AA,j,t} \int_1^{\infty} (W_{AA}(z\hat{q}^\eta, \epsilon', t+1)) dF_{j,t}(\hat{q}) \right. \\ \left. + (1 - \lambda) W_{AA}(z, \epsilon', t+1) \right) d\nu_{\epsilon'|\epsilon}$$

- Migrants from country A living in B

$$V_{AB}(z, \epsilon, t) = \pi_B(z + \epsilon, t) + \beta\delta \int_{-\infty}^{\infty} \left(\lambda \sum_j \psi_{AB,j,t} \int_1^{\infty} (W_{AB}(z\hat{q}^\eta, \epsilon', t+1)) dF_{j,t}(\hat{q}) \right. \\ \left. + (1 - \lambda) W_{AB}(z, \epsilon', t+1) \right) d\nu_{\epsilon'|\epsilon}$$

Model - BGP, Taxation and Migration Policies

- Balanced Growth Path Equilibrium
 - Agg productivity $\bar{A}_c$ grows at constant rate $g = g_A = g_B$ in each country
 $\implies$ constant productivity gap a
 - Stationary talent distribution ▶ BGP Def

Model - BGP, Taxation and Migration Policies

- Balanced Growth Path Equilibrium
 - Agg productivity $\bar{A}_c$ grows at constant rate $g = g_A = g_B$ in each country
 $\implies$ constant productivity gap a
 - Stationary talent distribution ▶ BGP Def
- Focus on EU-US migration $\rightarrow$ Country A = EU; Country B = US
- Country-specific proportional tax rate τ_c on inventor profits
 - Gov budget balances every period
 - Tax revenues fund lump-sum transfer to workers
 - $\tau_A > \tau_B$ $\tau_A = 0.4; \tau_B = 0.3$
- Migration restriction to country B: at most $\bar{\mu}$ migrant inventors from A can enter
 - If binding, random selection
 - Country A has no migration restriction

Model - Assumptions

Assumption 1:

- Same distributions of talent, productivity shock process, and initial share of inventors
 - i.e. $\theta_A = \theta_B$ $\rho_A = \rho_B$, $\omega_A = \omega_B$, $I_A = I_B$

Assumption 2: Compared with locals in A , migrants of nationality A are:

- (i) More likely to meet other migrants from A $\zeta_{AB,AB} > \zeta_{AA,AB}$
- (ii) More likely to meet locals in B $\zeta_{AB,BB} > \zeta_{AA,BB}$
- (iii) Less likely to meet migrants from B in A $\zeta_{AB,BA} < \zeta_{AA,BA}$

Similarly, for country B :

$$\zeta_{BA,BA} > \zeta_{BB,BA}, \quad \zeta_{BA,AA} > \zeta_{BB,BA}, \quad \text{and} \quad \zeta_{BA,AB} < \zeta_{BB,BA}.$$

Calibration

Table 5: Parameter Values

Parameter	Description	Value
— Panel A. External Calibration —		
β	Discount Rate	0.97
r	Interest Rate	0.03
δ	Survival Rate	0.95
α	Final Good Production	0.11
ν	Inventor-Firm match rate	1.00
τ_A	Tax Rate EU	0.40
τ_B	Tax Rate US	0.30
I_A	Share R&D workers	0.01
— Panel B. Direct Match to Data —		
$\xi_{AB,AA}$	Meeting Frictions	1.31
$\xi_{AB,BB}$	Meeting Frictions	0.65
$\xi_{BB,AA}$	Meeting Frictions	0.06
$\xi_{BA,AA}$	Meeting Frictions	0.71
$\xi_{BA,AB}$	Meeting Frictions	0.32
$\xi_{BA,BB}$	Meeting Frictions	1.24
— Panel C. SMM Calibration —		
$\bar{\mu}$	Migration cap to US (Share of Inventors)	0.01
κ	Cost of Migration	0.10
λ	Meeting Intensity HH	0.10
η	Learning Technology	0.34
σ	Technology Absorption	0.02
θ_A	Talent CDF H	15.00
ρ_A	Location Shock Persistence H	0.89
ω_A	Location Shock SD H	0.20

Note: List of model parameters and calibrated values. For the SMM calibration (Panel C), all parameters are calibrated jointly.

Results - Migration and Return Rules

- 3 reasons inventors choose to migrate:
 - i. Higher profits π
 - ii. Idiosyncratic productivity shocks ϵ
 - iii. Learning opportunities
 - Different $\psi_{i,j}$
 - High talent inventors learn more from peers
- Return to home country if foreign productivity ϵ falls sufficiently low

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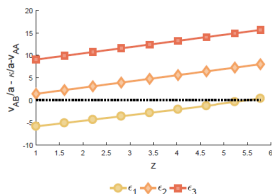
Proposition: Migration thresholds exist

- Along a BGP with assumptions 1 and 2, for every inventor type $j \in \{AA, AB, BB, BA\}$
 - $\exists$ talent $\bar{z}_j$ thresholds, given ϵ
 - $\exists$ productivity shock $\bar{\epsilon}_j$ thresholds, given z
 - Migrate if above threshold, return if below threshold

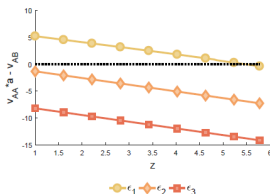
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BGP Migration and Return Decisions ($\epsilon_1 < \epsilon_2 < \epsilon_3$)

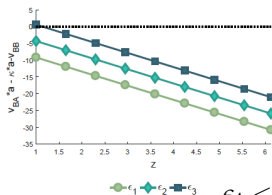
(a) Net Value of Migration for A Locals



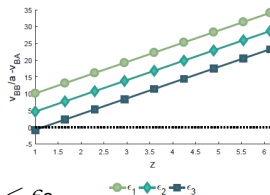
(b) Net Value of Return for A Migrants



(c) Net Value of Migration for B Locals



(d) Net Value of Return for B Migrants

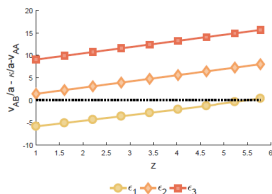


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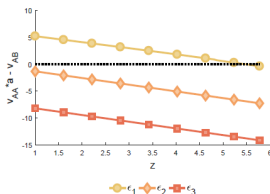
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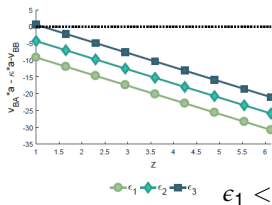
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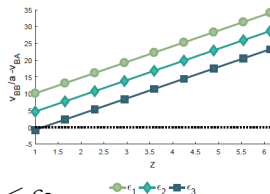
(b) Net Value of Return for A Migrants



(c) Net Value of Migration for B Locals



(d) Net Value of Return for B Migrants



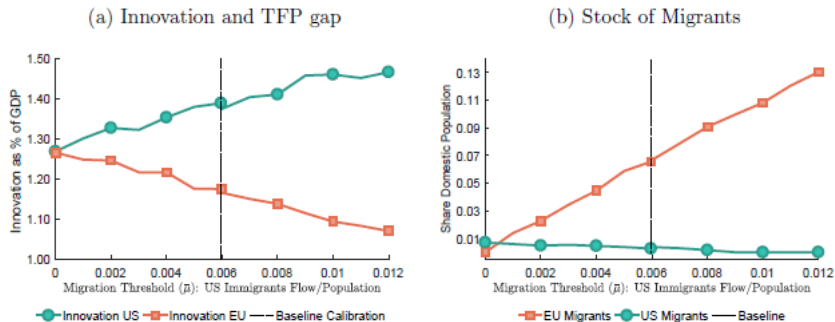
$\epsilon_1 < \epsilon_2 < \epsilon_3$

- In eq, country B has $\uparrow \bar{A}$, $\downarrow \tau$, and $\uparrow$ average talent z
- Positive slope from B to A: more talented ($\uparrow z$) inventors learn better from network
- Negative slope from A to B: more talented $\uparrow z$ inventors give up higher π and learning opportunities

Counterfactual: Increase Migration Cap $\bar{\mu}$ (BGP)

- $\uparrow \bar{\mu}$ leads to $\uparrow$ innovation in US and $\downarrow$ innovation in EU
 - Mainly b/c stock of migrants: increase of EU migrants to US, and fall of US migrants to EU

Figure 14: Counterfactual Change to US Immigration Threshold ($\bar{\mu}$): BGP Comparison



Increase Migration Cap $\bar{\mu}$ (Transition Dynamics)

- $\uparrow \bar{\mu}$ from 0.006 to 0.012

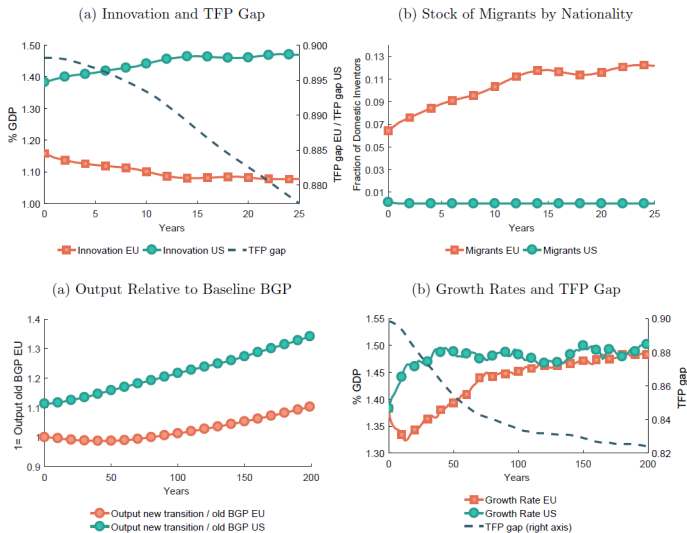
Short Run (15-25 years):

- $\uparrow$ brain drain from EU to US
- US innovation $\uparrow$ 7%, EU innovation $\downarrow$ 2%
- Productivity growth falls in EU by 0.05 p.p (4%)

Long Run:

- TFP growth $\uparrow$ in EU & US by 9%
- Technology diffusion and knowledge spillovers

Figure 16: Counterfactual Increase of US Migration Threshold: Transitional Dynamics.



Counterfactual: Tax Reduction $\downarrow \tau_A$

- $\downarrow \tau_A$ for foreigners and return migrants to EU $\tau_A = 0.30$

Counterfactual: Tax Reduction $\downarrow \tau_A$

- $\downarrow \tau_A$ for foreigners and return migrants to EU $\tau_A = 0.30$
- Short Run
 - Reduces net emigration from EU to US ($\downarrow$ brain drain)
 - Innovation $\uparrow$ EU by 9%, $\downarrow$ US by 6%
 - $\uparrow$ EU productivity growth by 5% in first 25 years
- Long run equilibrium
 - EU productivity growth falls by 6%
→ Less knowledge spillovers or technology diffusion
 - US productivity growth falls by 8%

▸ Transition Dynamics

▸ Channels

Conclusion

- This paper develops new model with innovation-endogenous growth
- Highlights importance of endogenous knowledge spillovers with networks of inventors
- Uses novel micro dataset of migrant inventors from patent data (EPO and PCT) to calibrate model
- Inventor migration is key for talent allocation, knowledge diffusion and productivity growth in the long run

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Extensions:

- Endogenous choice of occupation: worker or inventor
- Allow for migration of workers and depreciation of skill z

Appendix

Balanced Growth Path Equilibrium Definition

Definition 4 *Balanced Growth Path.* A BGP equilibrium consists of a constant growth rate g , a constant productivity gap a , and, for each country $c \in \{A, B\}$, paths for production workers wages $w_c(t)$, inventor profits $\pi_c(t)$, price of ideas $p_c(t)$, allocation of inventors across locations, $\mu_{AA}, \mu_{AB}, \mu_{BA}, \mu_{BA}$, productivity distributions $F_c(q)$ such that

1. The wage of production workers satisfies equation (2).
2. Profits of inventors satisfy equation (5).
3. Migration decisions are time invariant and solve equations (6),(7),(8), and (9).
4. The price of technology clears the market for ideas and satisfies equation (10).
5. The growth rate g and the productivity gap a satisfy equations (11) and (12).
6. Aggregate productivity $\bar{A}_c$ and aggregate output Y_c grow at rate g in each country.
7. The endogenous productivity distributions F_A and F_B are stationary, and the mass of individuals of each type $\mu_{AA}, \mu_{AB}, \mu_{BA}, \mu_{BA}$ is constant.

Counterfactual: Tax Reduction Transitional Dynamics

◀ Back

Figure 12: Tax Cut for Foreigners and Return Migrants in the EU: Transitional Dynamics.

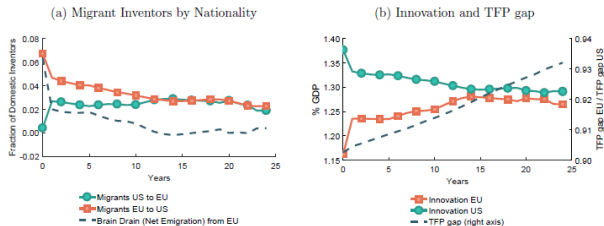
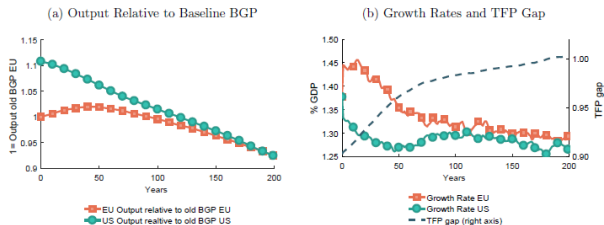


Figure 13: Tax Cut for Foreigners and Return Migrants in the EU: Transitional Dynamics.



Counterfactual: Tax Reduction Channels

◀ Back

Table 8: Tax Cut for Foreigners and Return Migrants in the EU: Effects on EU Output

Channel	Change in EU Output	
	After 25 years	After 200 Years
Direct Effect	+2.63	+32.50
Change in Return Migrants' Productivity	-0.36	-4.69
Knowledge Spillovers	-0.57	-9.77
Migrants' Selection	+0.65	+ 8.26
Change in Diffusion from US	-0.87	-33.77
Net Effect	+ 1.48	-7.47%

Notes: The table illustrates the change in EU output after 25 years and 200 years since a cut in the tax rate for foreigners and return migrants in the EU from 0.4 to 0.3. The table documents the separate impact of different channels and their net effect.

Thank You

Model

- OLG model in GE with innovation-based endogenous growth
- Discrete time
- 2 countries, $c \in \{A, B\}$
 - unit mass of indivs of each nationality
- 3 sectors: Final Good, Intermediary Goods, and Technology
 - Consumption of final good
 - New technology $\uparrow$ quality of intermediary goods and $\uparrow$ productivity growth
- Trade
 - Tradeable final good
 - Open capital markets (common exogenous interest rate r)
 - Closed technology sector